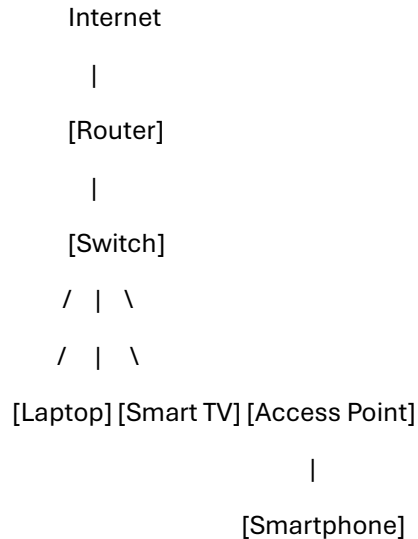


Network Devices

1. Simple Home Network Diagram



Connections

- Router → Connects home network to the internet.
- Switch → Connects multiple devices in the local network.
- Laptop → Connected to switch with Ethernet cable.
- Smart TV → Connected to switch with Ethernet cable.
- Smartphone → Connected wirelessly through the Access Point (Wi-Fi).

2. Comparison Table of Network Devices

Device	Purpose	Real-World Use Case
Router	Connects network to the internet	Using Wi-Fi to watch YouTube videos
Switch	Connects multiple devices in a LAN	Connecting PCs for LAN gaming
Hub	Broadcasts data to all devices	Small test network in a lab
Access Point	Provides wireless network access	Connecting smartphones and tablets to Wi-Fi

3. Scenario: Ordering Food on Zomato

When I order food on **Zomato** using my phone:

1. My phone sends the request through Wi-Fi to the **Access Point**.
2. The Access Point forwards the data to the **Switch**.

3. The Switch sends the data to the **Router**.
4. The Router forwards the request to the internet.
5. The request reaches Zomato's servers.
6. The response follows the same path back to my phone.

Roles

- **Access Point:** Provides Wi-Fi connection.
- **Switch:** Transfers data between local devices.
- **Router:** Connects the home network to the internet.

4. Connecting a Gaming PC Far from the Router

Option 1: Powerline Adapter

- One adapter plug into a wall socket near the router.
- Another adapter plugs into a wall socket near the PC.
- Data travels through the home's electrical wiring.
- The PC connects using an Ethernet cable.

Option 2: Wi-Fi Extender

- Place the Wi-Fi extender between the router and PC.
- The extender receives and boosts the Wi-Fi signal.
- Connect the PC to the extender using an Ethernet cable (if supported).

Result

Both options allow the gaming PC to access the internet even when it is far from the router.